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OncoHealth Interim Assessment

DRAFT - September 12, 2025

Newfire Project Team

Core Team

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Support Team

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Below are early hypotheses that the Newfire Advisory Services team are continuing to explore and refine. This serves as the starting point to shape the final recommendations and areas of investigation.

Overview

1. Project Updates
2. Early Findings
3. Risks, Goals, and Existential Requirements
4. Architectural Approach
5. Data Approach
6. Project Framework and Early Entries
7. Next Steps

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1 - Project Updates at Week 3

- Mostly completed first rounds of interviews. A couple Product team interviews remain
- Fewer interviews/interruptions moving forward, but we will need several to validate our hypotheses and insights.
- Completed first round of system mapping
    - First based on collaborative interviews
    - Backed by code exploration
- Good start on mapping APIs and Listing Services (needed for heatmaps)
- We will shortly add Inna and Bogdan - required for more detailed analysis on systems and code, respectively - to map the path forward.
- Completed analysis of source code change control. Initial findings below.
- Stress Testing an architectural proposal
- Initial rounds of Data systems interviews complete
- Data detailed reviews and planning will pick up on the back half of the engagement
    - Findings
    - Recommendations, Projects, to support above architecture, CS goals and Ops goals

2 - Early Findings

- A lot of what we know at this point is shared common knowledge, so we'll try to focus on what might be new, or what is relevant to project goals. Some relevant themes:
    - Staff is generally aggrieved
    - High commitment to near term goals but low sense of agency towards improving the situation
    - The situation is in a back-slide (all fronts, code complexity, tech debt, ops overhead)
    - Strategic initiatives are generally trumped by changes in direction, before they can complete.
    - Teams are structured based on type of work, not based on functional areas - everyone touches everything - with a lack of ownership - this breaks down
    - Too much work is done off platform (20-30% anecdotal)
    - Current focus is largely on symptoms, rather than cures

- Team structures handicap the benefit of Agile - PMs manage horizontal functions rather than vertical assets. Multiple teams managing the same code slows test and release cycles.
- OncoHealth's services are heavily supported by operations and clinical staff, with partial technical augmentation. Solutions to date have largely been technical MVPs that do not fully consider the manual processes.
- While Matis has a regimen-based approach and SmartUM has a drug-based approach, we have found both to be missing critical data to drive workflow. Client rules and clinical oversight requires drug-based details - down to formularies.
- Coding Hygiene is Poor. For the newer Monarch project the tech stack has improved but the code implementation is at roughly the same standards. Poor practices from Matis are reflected in Monarch. For example, much of the business logic is contained in a single 10,000-line JavaScript file in the .NET back-ends of both Matis and Monarch
- Matis is an over-worked monolith
    - >1,000,000 LOC. This is considerably larger than warranted for the scope of features.
    - ~84,500 commits over 8 years in approx 12,000 pull requests.
    - Anecdotally: For the last 5+ years most changes have been tactical/emergency.
    - There is no practical modularity in either the code or the data model, and we have not seen evidence of efforts to address technical debt - allowing it to accumulate significantly.
    - Implementing against the Matis monolith will cause a significant multiplier on efforts. This observation has been supported by the project scope of Admin Portal/Monarch.
- Additive approach to development (vs heavily indexing on refactoring). 30+ current active repositories/projects. 175 total. This is not necessarily bad, but supports the observation that there has been pretty consistent, and high level of tactical work for years.
- Additive approach to tools - Legacy tools are not generally retired.
    - For example, Mirth is still used and is causing drag. The team estimates the cleanup is a moderate to low effort, with high ROI. There isn't evidence of a product management channel for efforts like this.
- Fundamental software product capabilities are missing
    - Low-level logging and Observability
    - Code instrumentation
    - Data schema and elements (e.g., Basic drug-level data elements)
    - KPI Metrics (case workflow events and time stamps)
    - Centralized configuration management capabilities

- ASSERTION - Admin Portal / Monarch is unlikely to succeed as currently planned
    - Team structure and PM disciplines are unfavorable
    - Integrating with Matis creates huge R&D overhead (10x any new project)
    - Strategic initiatives are not firewalled, and strategy is in flux
    - Timelines are already too long - exposure to above is way too high.

Code Control Analysis (early)

Please reference Appendix A.

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3 - Risk, Goals, and Existential Requirements

Risks

- Compliance Risks from workflow and data gaps
- Back-sliding on Technical Debt
- Staff turnover from lack of clear ownership and strategic direction

Goals

- Isolate and Contain (From CritSit Playbooks)
- Define and execute strategic initiatives
    - Each as small as possible
    - Each addressing significant improvements

Existential Requirements

If these areas are not addressed then strategic initiatives are unlikely to succeed

- **Client Intake**
    - Defined and consistent process that accounts for scope of customizations
    - Separate Customer Implementation team from Strategic development teams.
        - Establish contingencies - if there are more sales and concurrent developments, then scale the Customer Implementation resources with contingent resources.
        - Clear ownership of Client success must be established, including implementations, and customization requests (separate and different from long-term product ownership).
    -
- **Product Management Leader**
    - Strong product head must be in place
        - Interim basis
        - Long term
    - Strategic Initiatives must be identified, staffed and protected, to prevent delays
    - We recommend the CPO (or equivalent) report to the CEO. We are generally supportive of the CPTO role, but in this case the need to separate strategic

priorities from implementation and client emergencies is paramount, and that has to be reflected in the org structure and ownership.

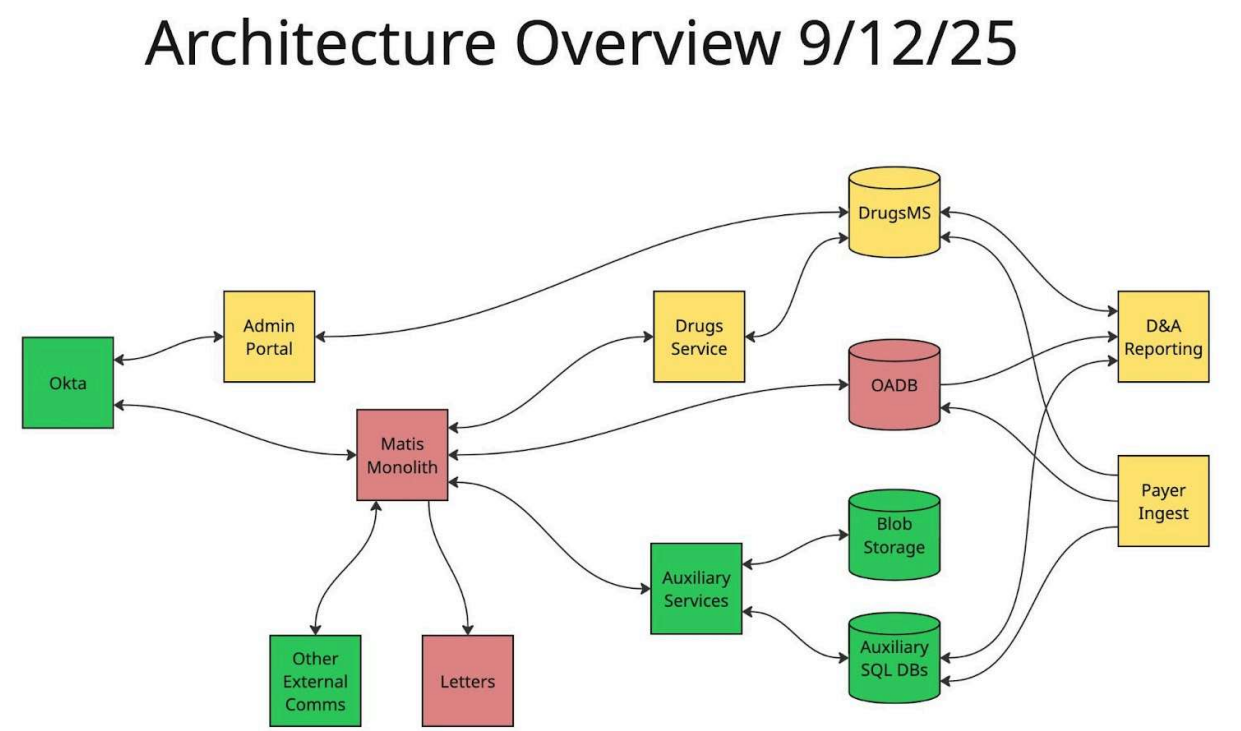
- **Product Development Reorg.**
    - There is no structural division of customer success teams and product feature development teams
    - Teams are defined loosely on implementation efforts, and generally not on areas of responsibility (Technical or Business).
    - A reorganization is required and procedural firewalls must be established to allow teams to reclaim ownership of their respective domains.
        - Client Implementations - with budget tied to ROI and margin funded by new-client sales. This includes Technical teams, client configuration, etc.
        - Product Development
            - New Feature Development - Including new customer requests, funded by customer SOWs and Business value/investment
            - Rebuilding efforts - modeling should be slightly different here - funds for rebuilding will have to be identified

- **Instrumentation and Observability**
    - Onco has critical blind spots to self-measuring performance
        - Audit trails have to be implemented on cases, including understanding time spent in respective parts of the workflow, and dispositions. Helpful for debugging, reporting and compliance. Production audit trail must be PHI-free for offshore dev. access.
        - Proper systemic logging/debugging must be in place. This can start with implementation of Observability tools like New Relic or DataDog.

4 - Architectural Approach

Recommended approach to managing the many changes that are needed to scale the business and to address years of accumulated technical debt found in the code, business logic, and data.

Assertions



- Matis Monolith (Core) is no longer viable. (see above observations) It's the source of many downstream failures. Matis Monolith and the central DB require a careful but urgent replacement to unlock the ability to address tech debt and to consolidate the client base on a single platform.
- This does not include the many other projects/services - most of which are good enough for now, but will benefit from future refactoring AFTER Matis services and Database are restructured.

High Level Approach

1. Create a duplicate New OneUM deployment based on Matis, leaving the Legacy OneUM deployments in place.
    a. The business and current initiatives must not be interrupted - and continue on Legacy.
    b. Share common, stateless, services across the two deployments where possible (instead of duplicating). Duplicate the remaining (but do not recode).
    c. Place a facade on Matis monolith (prime) and reimplement with new modular services including:
        i. Configuration Management
        ii. Case Workflow Modeling
        iii. Full Drug-level and Formulary Modeling
        iv. Full Regimen/Protocol Modeling
        v. Provisions for Provider and Unified Member
        vi. Support for Admin Portal (no separate follow-on project)
2. Product Designs, Plans, Architecture in place for a multi-phase rewrite of the core schema and core SET of replacement services
    a. Scope is not known yet, but should be a fraction of the total Admin Portal/Monarch Project (on cost and time).
    b. Team structure, executive support and support roles will be critical to success
3. Follow with a comprehensive data strategy, including least-impact possible to migrating existing clients (combination of planning, tools, and clinical policy and operations support)
4. Ongoing - Run several improvement projects in parallel, replacing/upgrading systems to take advantage of new OneUM core services.
5. Existing Matis clients, new clients and SmartUM clients all migrate
6. Shut down SmartUM and Legacy Matis once clients are off.

5 - Data Approach

Data analysis efforts have started, and will ramp up in weeks 5-8 of this engagement. These will include analysis of gaps and tech debt in the context of the architectural approach above, as well as product initiatives (below) that can help with standardizing interoperability (FHIR) and the opportunity to leverage AI capabilities in data ingest, as well as in dispositioning prior approvals - helping reduce/shift human processes.

6 - Project Framework and Early Entries

Many areas are known to have positive ROI if they can be funded/staffed in time. This assumes structural changes (above) are in place and effective. The following classification mechanism forms the basis of a rubric for helping manage additional improvements to the system, while a heart-transplant is under way.

Goals

- Address Drag from Technical Debt
- Address Drag from manual Operational and Clinical Processes
- Address Risks to the business

Some Example Projects by Classification

- **Continued Legacy** Projects - made in the legacy code base, on a required basis, with modest improvements to methodology. Enables continued improvements and implementations.
    -
- **Net New** Capabilities - projects that can be implemented in both the legacy and new implementations.
    - Code Instrumentation and system Observability - cost effective and required
    - Depreciate and Cleanup Mirth
    - Letters v2 (in progress) - Under way - reasonable tech - still evaluating architecture and code quality - good business ROI.
    - Turnaround Time - Well built so far. Fairly good ROI - Under way. Determine ongoing prioritization
    - PHI-Free Audit/Logging
- **Cleanup and Recode** - projects that address configurability and features of the new system, managed with strict configuration capabilities. Evaluated on a cost/benefit prioritization basis.
    - Configurable Case Workflows, w/ modular events, rules and triggers. High cost with high ROI and required.
    - Case module/API (new to Matis)
    - Drug-level Statusing

- New **Features** - extensions of the NEW implementation only, made to new standards. Different from Net New in that they require the new data schema and service endpoints
    - Provider Mastering
    - Global Member Mastering

7 - Next Steps

- Continue code level assessments (.Net engineer) and feature-level documentation (Business Analyst) - with a lens on above goals
- Refine and validate proposal
- Complete data systems assessments and generate recommendations
- Continue to document discrete projects and complete rubric for PM evaluation
- Complete final report for SOW 1

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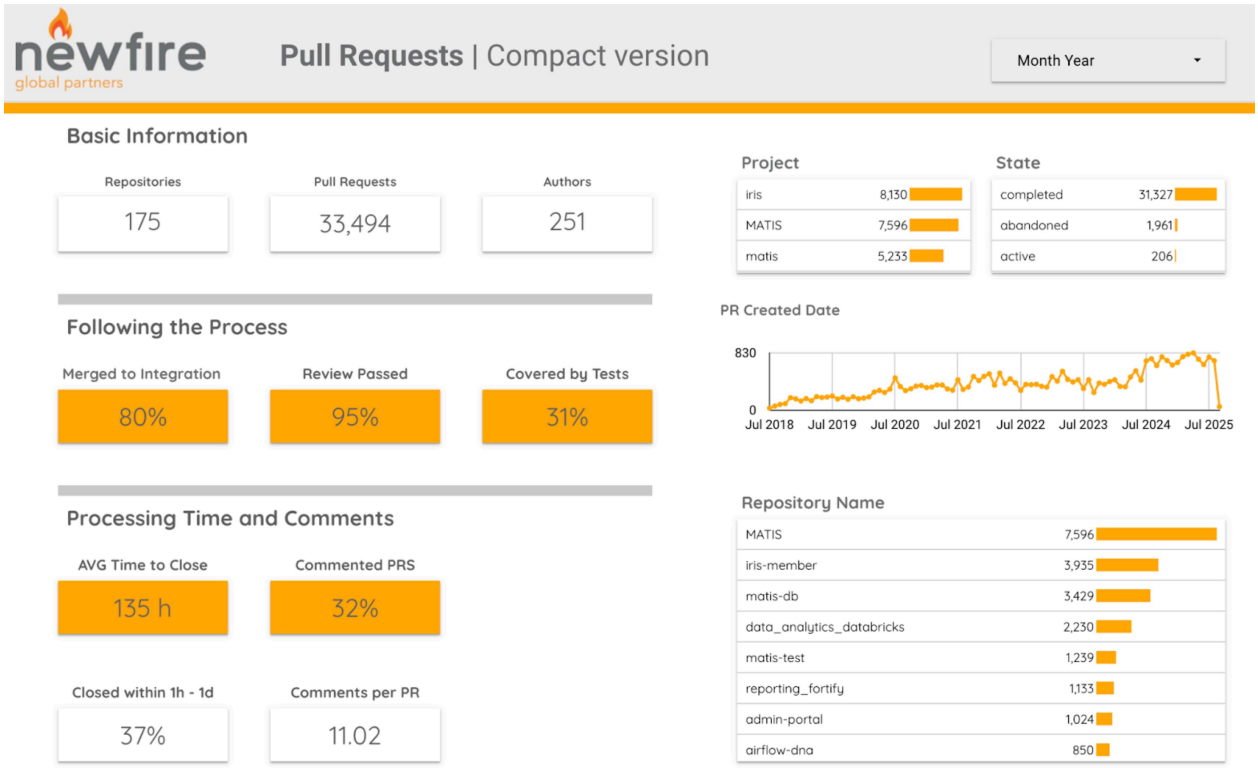
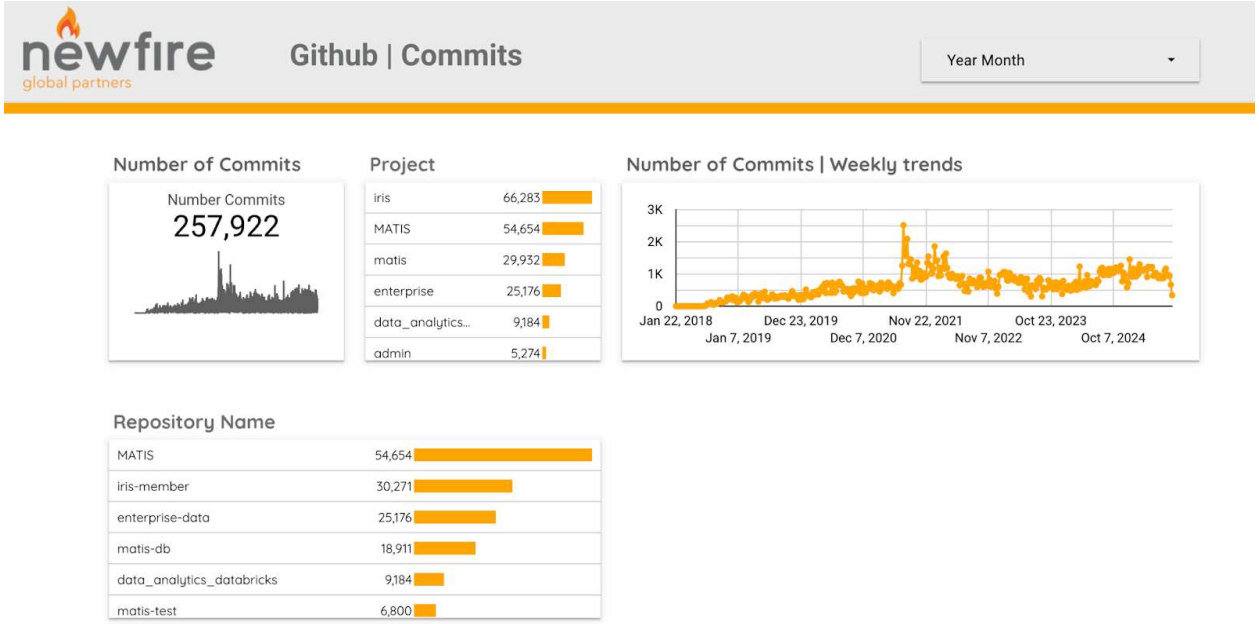
APPENDIX A

Pull Request Analysis

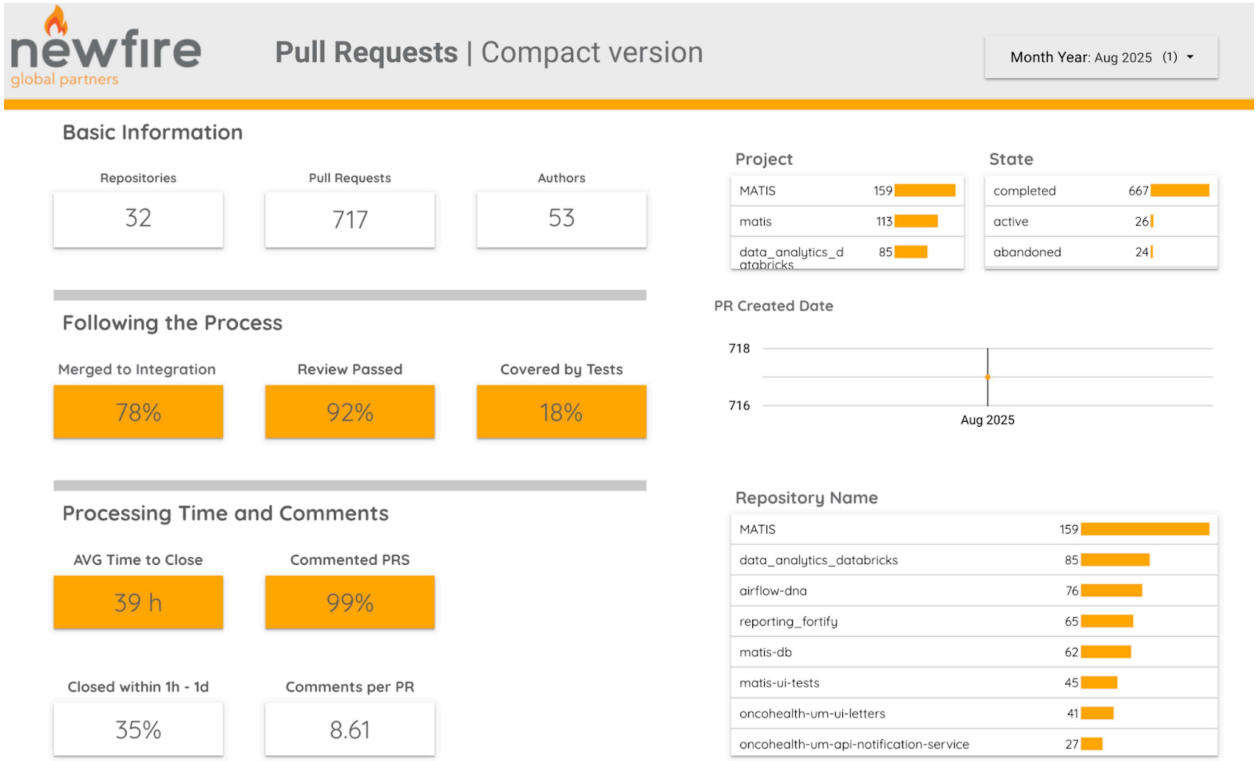
All-Time PRs



All-Time Commits



Pull-Requests in August



- PRs Hygiene has been good the last several months (almost all PRs have comments). Historically that has not been the case.

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Preliminary Statistics by Team - Last 12 Months

Team	Average of Commits	Average of PRs	Average of % of PRs with Comments
Engineering - Aurora (client implementations)	816	147	23.64%
Engineering - Dragons (strategic development)	1068	206	54.24%
Engineering - Everest (efficiency projects)	732	77	56.54%
Engineering - Genesis	0	0	0
Engineering - Iris	1681	287	87.71%
Engineering - Monarch (Matis future state)	1231	138	44.98%
Engineering - Phoenix (production support)	770	175	28.59%
**Filtered to software engineers only (TPMs, BA, QA excluded)			

By Employer (last year)

Employer	Average of Commits	Average of PRs	Average of % of PRs with Comments
Onco	827	159	**60.2%**
SS	956	160	43.4%
Versa	1130	143	41.1%
**Filtered to software engineers only (TPMs, BA, QA excluded)			

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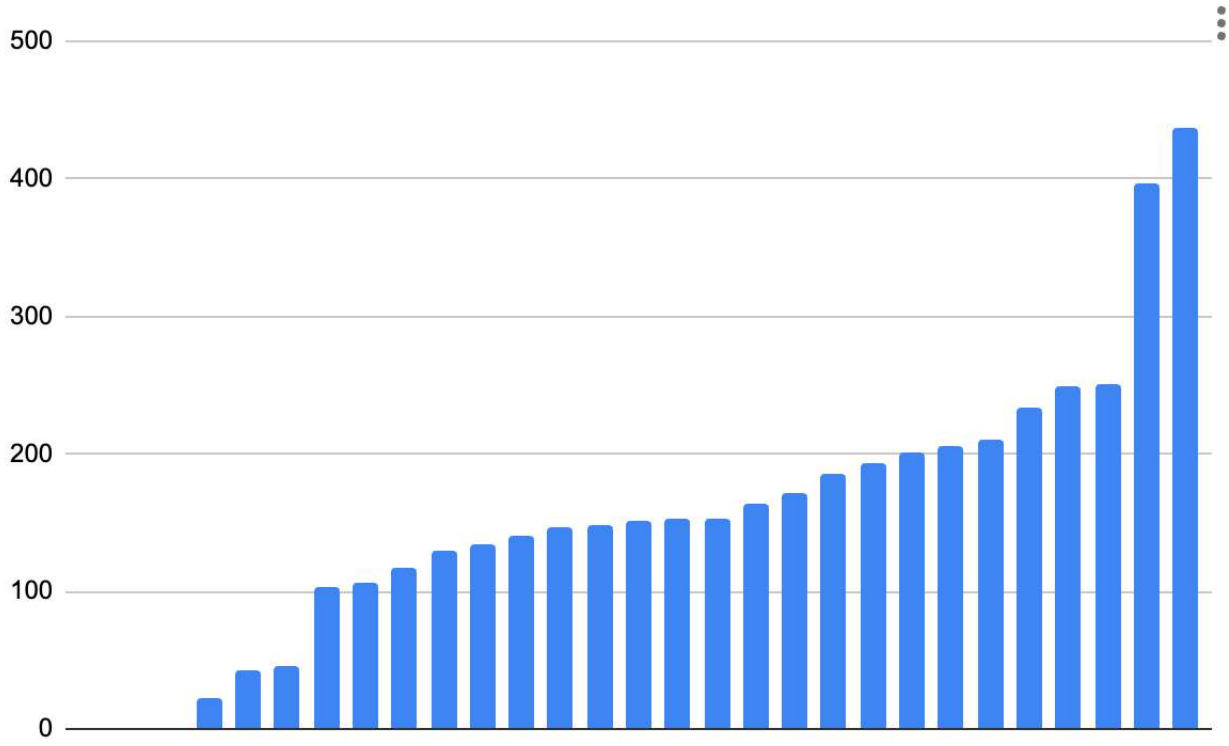
Number of PRs for Each Software Developer (Last 12 Months)